

2024



Nearmap Uses Amazon SageMaker to Scale Machine Learning of Complex Computer Vision Analytics

Learn how location-intelligence provider Nearmap improved its data science management and AI capabilities on AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Streamlines

model training processes

Reduces

operating costs

Facilitates

in-depth analytics of Nearmap aerial imagery

Supports

sustainability initiatives with data



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50+ PB of data stored

on Amazon S3

Overview

With its aerial camera technology, [Nearmap](#) offers organizations a dynamic lens to track structural and environmental changes over time. Taking full advantage of over 50 PB of imagery data with custom deep learning models requires advanced and powerful technology.

To support its growing machine learning (ML) training and artificial intelligence (AI) needs, Nearmap upgraded from on-premises hardware to robust and scalable solutions from Amazon Web Services (AWS). Large, custom deep learning models, and the current trend toward large vision models, require the ability to dynamically scale up training that uses multiple machines at once. Nearmap not only found a way to reduce the time to train an ML model through larger bursts of training compute but has also paved the way for even larger models that unlock the full value of its dataset, revolutionizing its AI capabilities.



Opportunity | Using AWS to Support Advanced Computer Vision Use Cases

Founded in 2008, Nearmap uses aerial camera technology and in-house vision and ML pipelines to help organizations track changes, such as urban growth, the effects of natural disasters, and changes in tree coverage. It serves a diverse array of customers, from property insurance providers to government entities and disaster relief organizations.

“We capture huge amounts of aerial imagery and turn those images into 2D visual maps and 3D models of whole cities,” says Dr. Michael Bewley, vice president of AI and computer vision at Nearmap. “We map the world, and it’s constantly changing over time, not only visually, but with AI.”



To support its ML and AI needs, Nearmap worked with a local cloud provider and reserved a series of A100 GPUs for model training, which required careful planning and coordination between teams to reserve enough capacity and optimize usage. However, Nearmap found that it required more flexibility and scalability to train its ML models on multiple nodes of A100 GPUs and reduce the need to plan for compute availability. After exploring potential solutions, Nearmap set its sights on AWS.

“On AWS, we’re free from managing overlapping resources, which simplifies planning for project and model training,” says Dr. Nagita Mehr Seresht, senior director of AI model research and development (R&D) at Nearmap.

Solution | Training ML Models to Unlock over 50 PB of Imagery Data Using Amazon SageMaker

Nearmap adopted the AWS service [Amazon SageMaker](#)—a fully managed service that helps developers and data scientists build, train, and deploy ML models for nearly any use case—and uses [Amazon Simple Storage Service](#) (Amazon S3), an object storage service built to retrieve any amount of data from anywhere. By using these services in tandem, Nearmap can perform advanced analytics at scale and with less operational overhead. “Using Amazon SageMaker, our model training process has become much more streamlined, which helps us perform ML at scale,” says Dr. Seresht.

Previously, Nearmap’s model training required local data syncing to GPUs, which was time intensive. With Amazon SageMaker Model Training capabilities, the company no longer needs to perform this cumbersome process; Nearmap can train models directly on Amazon SageMaker using data stored in Amazon S3, leading to faster model training and deployment. To perform all its serious vision model training, the company relies on [Amazon SageMaker Model Training](#), which reduces the time and cost to train and tune ML models at scale without the need to manage infrastructure. Nearmap can also perform multiple model training tasks in parallel on Amazon SageMaker, which further accelerates the ML training process.

Given the immense volume of aerial imagery that Nearmap possesses—totaling over 50 PB—the company needed reliable and scalable storage. With Amazon S3, the company can access, store, and manage data as needed. “By directly reading from and writing to Amazon S3, we have virtually infinite storage,” says Dr. Seresht. “Additionally, the security, maintenance, and overall management are taken care of, which is a significant advantage.”

Using AWS services, Nearmap’s team members have access to the data and ML tools that they need to experiment with AI and ML. They can carry out these projects without encountering resource constraints, which accelerates the pace of innovation. “On AWS, each person can decide how many experiments they want to run in parallel,” says Dr.



Seresht. “They might run multiple experiments for a few days and not do any training for the next week, and we can accommodate that on AWS.”

On AWS, Nearmap has the resources and the capabilities that it needs to explore in-depth analytics projects. “Typically, when such studies are done, they’re limited to a town or city, using tools such as custom-built ML models tailored for a single customer,” says Dr. Bewley. “In contrast, we use a consistent deep learning model and software across multiple cities in Australia and the United States. As a result, we can make valid comparisons not only within a city but also between cities and even countries.”

By understanding the distribution of tree canopies in dense urban areas, Nearmap helps policymakers make informed decisions regarding urban planning and reforestation efforts. “Using our extensive data on AWS, I demonstrated that Adelaide, Australia, for example, lost 9.8 percent of its relative tree cover over a decade,” says Dr. Bewley. “We can do this for any city covered by our imagery capture program. This level of detail and historical comparison was previously unattainable.”

Outcome | Harnessing Insights to Promote a Cleaner, More Sustainable World

By adopting Amazon SageMaker, Nearmap can access and analyze petabytes of information for AI and ML. The company’s data is growing every day, and the underlying computer resources of Amazon SageMaker help Nearmap scale seamlessly to accommodate the demands of its business, team members, and customers.

In the future, Nearmap wants to delve even deeper into data analysis—and AWS will continue to play an integral role in its technology stack. “Our ambition is to be the source of truth that shapes our livable world,” says Dr. Bewley. “We want to provide that layer of insight for people to understand how our cities are growing and evolving. By establishing a clear source of truth on AWS, our people and our customers can effectively use our data and focus on the crucial task of advocating for a cleaner, more sustainable future.”

About Nearmap

Founded in 2008, Nearmap is a location-intelligence company that uses aerial camera technology to help organizations in Australia, New Zealand, the United States, and Canada track changes in their environments.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is built on Amazon’s two decades of experience developing real-world ML applications, including product recommendations, personalization, intelligent shopping, robotics, and voice-assisted devices.

[Learn more »](#)



Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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Amazon SageMaker Model Training

Amazon SageMaker Model Training reduces the time and cost to train and tune machine learning (ML) models at scale without the need to manage infrastructure.

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2020



General Electric on AWS

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2015

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